

Yonghao Shi

Ph.D. Student in Computer Science, Simon Fraser University

Human-Computer Interaction | Smart Environments | Computational Materials | Applied AI

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Research Profile

I study human-computer interaction systems that connect smart materials, room-scale sensing, fabrication, and machine learning. My recent work explores battery-free wood panels, RFID-augmented environments, vibrotactile rendering, smart furniture, and low-barrier prototyping platforms for interactive devices.

Education

2022-now	Ph.D. in Computer Science, Simon Fraser University, Burnaby, Canada. Thesis: Human-Computer Interaction: Smart Environment and Computational Material.
2017-2018	M.Sc. in Computer Science, University of Plymouth, Devon, United Kingdom. Thesis: Data Science and Business Analysis.
2016-2017	B.Sc. in Electrical and Electronic Engineering, University of Plymouth, Devon, United Kingdom.
2013-2016	B.Sc. in Electrical and Electronic Engineering, Beihang University, China.

Employment History

2020-2021	Project Manager, Institute of Information Engineering, Chinese Academy of Sciences.
2018-2020	Software Engineer, Institute of Information Engineering, Chinese Academy of Sciences.
2017.6-2017.9	Software Engineer, PwC Beijing.

Peer-Reviewed Publications

SCF 2025	3Duino: A Low-Barrier Platform for Prototyping Interactive 3D-Printed Devices. Yonghao Shi, Zejia Cai, Liuyi Chen, Yuning Su, and collaborators. ACM Symposium on Computational Fabrication, 2025. DOI: 10.1145/3745778.3766649.
CHI 2024	WooDowel: Electrode Isolation for Electromagnetic Shielding in Triboelectric Plywood Sensors. Yonghao Shi, Chenzheng Li, Yuning Su, Xing-Dong Yang, and Te-Yen Wu. Proceedings of the CHI Conference on Human Factors in Computing Systems, 2024. DOI: 10.1145/3613904.3642304.
CHI 2024	Tagnoo: Enabling Smart Room-Scale Environments with RFID-Augmented Plywood. Yuning Su, Tingyu Zhang, Jiuen Feng, Yonghao Shi, Xing-Dong Yang, and Te-Yen Wu. Proceedings of the CHI Conference on Human Factors in Computing Systems, 2024. DOI: 10.1145/3613904.3642356.
IMWUT 2023	Laser-Powered Vibrotactile Rendering. Yuning Su, Yonghao Shi, Da-Yuan Huang, and Xing-Dong Yang. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 2023. DOI: 10.1145/3631449.

Current Manuscripts and Recent Submissions

Note: entries with ellipses use the short titles available in the submission record.

Apr 03, 2026	Lumberina: A Wireless Smart Wood Panel with Concealed Dot-Matrix Display ? GI 2026 Papers Round 2 (April), complete.
Mar 31, 2026	EcoVibWood: Sustainable ... ? UIST 2026 Papers, complete.
Jan 30, 2026	Development and Evaluation of ... ? GI 2026 Papers Round 1 (January), complete; final version available.
Jan 30, 2026	Lumberina: A Wireless Smart Wood Panel with Concealed Dot-Matrix Display ? GI 2026 Papers Round 1 (January), complete.
Dec 04, 2025	Development and Evaluation of ... ? CHI 2026 Papers, complete.
Dec 04, 2025	Lumberina: A Battery-Free, Wireless Smart Wood Panel with Concealed Dot-Matrix Display ? CHI 2026 Papers, complete.

Submission History

Aug 01, 2025	AccelLumber: Battery-free Smart Wood Panels with an Integrated Accelerometer for Activity Sensing on Modular Furniture ? IMWUT 2025 February cycle, not accepted.
May 16, 2025	CSCW 2025 Doctoral Consortium submission ? reject.
Apr 09, 2025	3Duino: A Low-Barrier Platform for Prototyping Interactive 3D-Printed Devices ? UIST 2025 Papers, not accepted.
Apr 09, 2025	FurniAntaeus: Floor-Powered Smart Furniture ? UIST 2025 Papers, not accepted.
Apr 09, 2025	Lumberina: A Battery-Free, Wireless Smart Wood Panel with Concealed Dot-Matrix Display ? UIST 2025 Papers, not accepted.
Jan 20, 2025	3Duino: A Toolkit for ... ? DIS 2025 Papers, not accepted.
Dec 10, 2024	3Duino: A Toolkit for ... ? CHI 2025 Papers, not accepted.
Dec 10, 2024	Lumberina: Wireless ... ? CHI 2025 Papers, not accepted.
Dec 10, 2024	AccelLumber: Battery-free ... ? CHI 2025 Papers, not accepted.
Apr 03, 2024	Lumberina: Wireless ... ? UIST 2024 Papers, not accepted.
Dec 12, 2023	WooDowel and Tagnoo ? CHI 2024 Papers, accepted.
May 15, 2023	Laser-Powered Vibrotactile Rendering ? IMWUT 2023 May cycle, accepted for publication.

Skills

Coding	Java, C, Python, R, SQL, XML/XSL, LaTeX.
Databases	MySQL, Oracle, SQL Server, Redis, MongoDB, Neo4j.
Research	Academic research, teaching, training, consultation, HCI prototyping, data analysis.

Updated April 2026.